

Fig. 4: Actual-size, single-side PCB layout for the humidity indicator and controller circuit

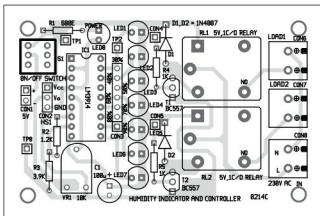


Fig. 5: Component layout for the PCB

transduction scheme that they use to convert water vapour concentration into an electrical signal, viz, capacitive, resistive (DC resistance or AC impedance) and piezo-resistive.

The SY-HS-220 series humidity sensor is based on capacitive technology. This sensor module converts the RH directly into equivalent millivolts analogue output. The driver circuit is in-built in the sensor assembly (see Fig. 2). The rated voltage is 5V DC. Its operating temperature range is 0-60°C and humidity is 30-90 per cent RH. Its standard output at 25°C and 60 per cent RH is 1980mV DC.

Fig. 3 shows the millivolts output reading corresponding to RH. The output of humidity sensor HS1 is connected to analogue input pin 5 of LM3914 through preset of 10-kilo-ohm (VR1).

Dot/bar display driver (LM3914). The LM3914 is an 18-pin DIP package IC that accepts analogue signals at its input pin 4 and drives 10 LEDs, providing a linear analogue display. Pin 9 is the mode-select input control for bar- or dot-mode operation. LM3914 works in bar mode when pin 9 is connected directly to pin 3 and pulled to +5V supply.

At any point of time, if all LEDs (LED1 through LED7) glow, it indicates that the humidity level is 90 per cent RH and if only LED1 glows, it indicates that humidity is at 30 per cent RH. The number of glowing LEDs increases with increase in humidity level

as shown in Fig. 1.

Before using this circuit, humidity level is first calibrated using preset VR1. You may refer to a standard humidity meter to calibrate it. For example, if a standard humidity meter shows 40 per cent RH, LED1 and LED2 should glow. If not, vary VR1 till LED1 and LED2 glow. Once calibration is done, you should not touch the preset again. Thus, if humidity level is 50 per cent, you should see LED1, LED2 and LED3 glowing.

T1 and T2 are pnp transistors for controlling two electrical devices through relays RL1 and RL2. You can trigger these relays at the desired humidity level by connecting their bases to the corresponding pins of IC1. All pins are terminated at CON3 for ease of connection. Note that RL2 can be used to stop the current process when predetermined humidity level is reached.

Construction and testing

An actual size, single-side PCB for the humidity indicator and controller is shown in Fig. 4 and its component layout in Fig. 5. Assemble the components on the PCB. Before switching on the circuit, check proper connections, especially correct polarity of the power supply at CON1. Normally, you will find humidity sensor module with five pins. You need to use only three pins (Vcc, Vo and GND) of this module, which are to be connected at CON2 in this circuit.

PARTS LIST

Semiconductors:

- IC1 - LM3914 dot/bar display driver
- T1, T2 - BC557 pnp transistors
- D1, D2 - 1N4007 rectifier diode
- HS1 - SY-HS-220 humidity sensor module
- LED1-LED8 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 680-ohm
- R2 - 1.2-kilo-ohm
- R3 - 3.9-kilo-ohm
- R4, R5 - 1-kilo-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1 - 100 μ F, 16V electrolytic

Miscellaneous:

- S1 - On/off switch
- CON1 - 2-pin Berg Strip connector
- CON2 - 3-pin Berg Strip connector
- CON3 - 7-pin Berg Strip connector
- CON4, CON5 - Jumper wire
- CON6-CON8 - 2-pin, 5mm terminal connector
- REL1, REL2 - 18-pin IC base
- REL1, REL2 - 5V, 1C/O relay

When circuit is turned on, vary VR1 till all the LEDs glow, to check if they are working well along with the driver.

For troubleshooting, you can refer the test points given in the table. Now, calibrate the circuit using a standard humidity meter as explained earlier. For manual testing, you can blow air from your mouth on the humidity sensor (HS1) and observe the LEDs glowing. The number of glowing LEDs depends on the moisture content in your breath. ●

The author is a B.Tech in electronics and communications engineering from Punjab Technical University